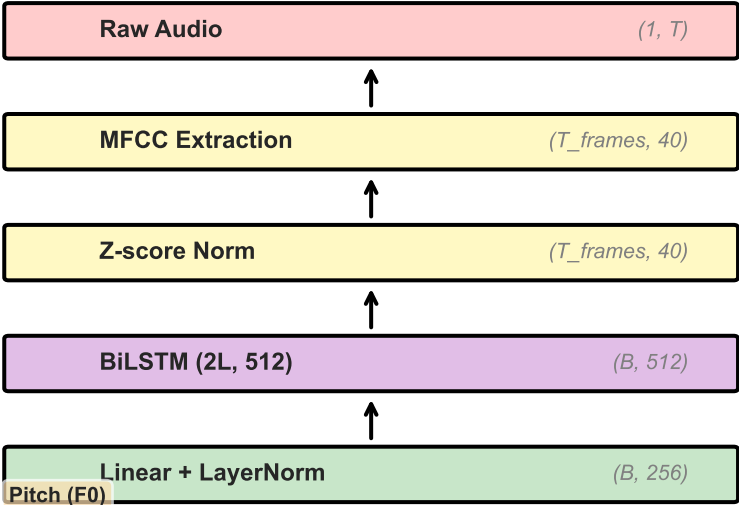


Speech Encoder Architecture Details

MFCC Path



Pitch (F0)
Formula: $ZCR * sr / (2 * frame_len)$
Depression: Flat pitch ↓ variation

Energy
Formula: $mean(RMS[voiced])$
Depression: Lower energy ↓

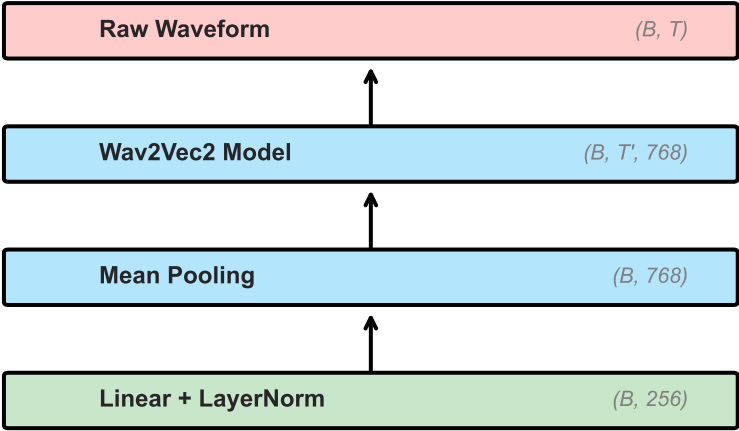
Response Latency
Formula: $first_voiced / n_frames$
Depression: Delayed onset ↑

Prosodic Features Extraction

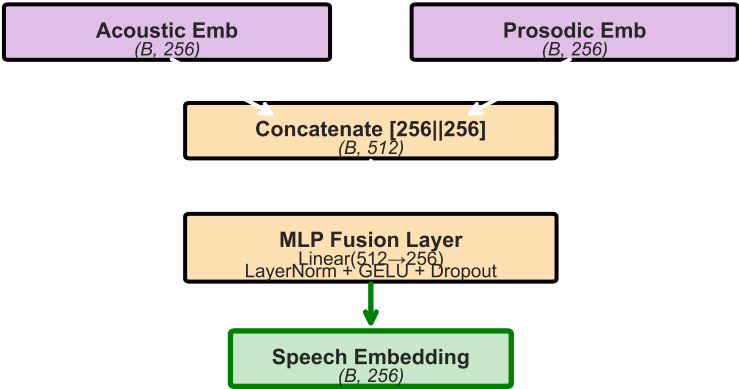
Pause Duration
Formula: $mean(silence_segments)$
Depression: Longer pauses ↑

Speech Rate
Formula: $voiced_frames / total_frames$
Depression: Rate ↓ in depression

Wav2Vec2 Path



Acoustic + Prosodic Fusion



Equation:

$$out = MLP(concat([acoustic, prosodic]))$$